

```
(%i6) /* Symbols and functions */
declare(x, real)$;
declare([x_n, v_n, h], [real, constant])$;
declare(m, integer)$;
assume(m > 0)$;
assume(h > 0)$;
depends([y, v], x)$;
```

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(%i8) /* Taylor expansion y(x_n + h)*/
```

```
taylor(y(x), x, x_n, 5)$;
```

```
y_plus: subst(x - x_n = h, %);
```

$$y_{\text{plus}} = \frac{h^5 \left(\frac{d^5}{dx^5} y(x) \right) \Big|_{x=x_n}}{120} + \frac{h^4 \left(\frac{d^4}{dx^4} y(x) \right) \Big|_{x=x_n}}{24} + \frac{h^3 \left(\frac{d^3}{dx^3} y(x) \right) \Big|_{x=x_n}}{6} + \frac{h^2 \left(\frac{d^2}{dx^2} y(x) \right) \Big|_{x=x_n}}{2} + h \left(\frac{d}{dx} y(x) \right) \Big|_{x=x_n} + y(x_n)$$

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(%i9) subs: [ /* System: y'(x) = v(x) and v'(x) = 2 x v(x) - 2 m y(x) */
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```
at(diff(y(x), x, 5), x = x_n) = at(diff(2 · x · v(x) - 2 · m · y(x), x, 3), x = x_n),
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at(diff(y(x), x, 4), x = x_n) = at(diff(2 · x · v(x) - 2 · m · y(x), x, 2), x = x_n),
```

```
at(diff(y(x), x, 3), x = x_n) = at(diff(2 · x · v(x) - 2 · m · y(x), x, 1), x = x_n),
```

```
at(diff(y(x), x, 2), x = x_n) = at(2 · x · v(x) - 2 · m · y(x), x = x_n),
```

```
at(diff(y(x), x, 1), x = x_n) = at(v(x), x = x_n),
```

```
at(diff(v(x), x, 5), x = x_n) = at(diff(2 · x · v(x) - 2 · m · y(x), x, 4), x = x_n),
```

```
at(diff(v(x), x, 4), x = x_n) = at(diff(2 · x · v(x) - 2 · m · y(x), x, 3), x = x_n),
```

```
at(diff(v(x), x, 3), x = x_n) = at(diff(2 · x · v(x) - 2 · m · y(x), x, 2), x = x_n),
```

```
at(diff(v(x), x, 2), x = x_n) = at(diff(2 · x · v(x) - 2 · m · y(x), x, 1), x = x_n),
```

```
at(diff(v(x), x, 1), x = x_n) = at(2 · x · v(x) - 2 · m · y(x), x = x_n)
```

```
];
```

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(%i11) /* Apply the system */
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y_plus: subst(subs, y_plus), nouns$;
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```
y_plus: subst(subs, y_plus), nouns$;
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(%i12) /* y_{n+1} = ... */
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```
y_plus: map(factor, collectterms(y_plus, h));
```

$$y_{\text{plus}} = \frac{h^5 (4 x_n^4 y(x_n)^2 m^2 + v(x_n)^2 m^2 - 4 x_n^3 y(x_n) m - 10 x_n^2 y(x_n) m - 6 x_n^2 v(x_n) m - 4 v(x_n) m + 4 x_n^4 v(x_n) + 12 x_n^2 v(x_n) + 3 v(x_n)) / 30 + h^4 (y(x_n)^2 m^2 - 2 x_n^2 y(x_n) m - 2 y(x_n) m - 2 x_n v(x_n) m + 2 x_n^3 v(x_n) + 3 x_n v(x_n))}{6} - \frac{h^3 (2 x_n^3 y(x_n) m + v(x_n) m - 2 x_n^2 v(x_n) - v(x_n))}{3} - h^2 (y(x_n) m - x_n v(x_n)) + y(x_n) + h v(x_n)$$

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(%i14) /* Same procedure for v(x) */
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```
taylor(v(x), x, x_n, 5)$;
```

```
v_plus: subst(x - x_n = h, %);
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$$v_{\text{plus}} = \frac{h^5 \left(\frac{d^5}{dx^5} v(x) \right) \Big|_{x=x_n}}{120} + \frac{h^4 \left(\frac{d^4}{dx^4} v(x) \right) \Big|_{x=x_n}}{24} + \frac{h^3 \left(\frac{d^3}{dx^3} v(x) \right) \Big|_{x=x_n}}{6} + \frac{h^2 \left(\frac{d^2}{dx^2} v(x) \right) \Big|_{x=x_n}}{2} + h \left(\frac{d}{dx} v(x) \right) \Big|_{x=x_n} + v(x_n)$$

$$\left(\frac{d}{dx} v(x) \right) \Big|_{x=x_n} + v(x_n)$$

(%i17) **v_plus: subst(subs, v_plus), nouns\$;**
v_plus: subst(subs, v_plus), nouns\$;
v_plus: subst(subs, v_plus), nouns\$;

(%i18) /* v_{n + 1} = ... */
v_plus: map(factor, collectterms(v_plus, h));

$$\begin{aligned} & - (h^5 (y(x_n)^3 m^3 - 6 x_n^2 y(x_n)^2 m^2 - 6 y(x_n)^2 m^2 - 3 x_n v(x_n)^2 m^2 + 4 x_n^4 y(x_n) m + 18 x_n^2 y(x_n) m \\ & + 8 y(x_n) m + 8 x_n^3 v(x_n) m + 15 x_n v(x_n) m - 4 x_n^5 v(x_n) - 20 x_n^3 v(x_n) - 15 x_n v(x_n)) / 15) + h^4 \\ & (4 x_n y(x_n)^2 m^2 + v(x_n)^2 m^2 - 4 x_n^3 y(x_n) m - 10 x_n y(x_n) m - 6 x_n^2 v(x_n) m - 4 v(x_n) m + 4 x_n^4 \\ & v(x_n) + 12 x_n^2 v(x_n) + 3 v(x_n)) / 6 + \\ & \frac{2 h^3 (y(x_n)^2 m^2 - 2 x_n^2 y(x_n) m - 2 y(x_n) m - 2 x_n v(x_n) m + 2 x_n^3 v(x_n) + 3 x_n v(x_n))}{3} - h^2 \\ & (2 x_n y(x_n) m + v(x_n) m - 2 x_n^2 v(x_n) - v(x_n)) - 2 h (y(x_n) m - x_n v(x_n)) + v(x_n) \end{aligned}$$